

Retrieval-Augmented Generation, Briefly

Overview

Retrieval-Augmented Generation, or RAG, is a technique for making a language model's answers grounded in a specific set of documents instead of relying only on what it memorized during training. Rather than fine-tuning a model on new data, RAG fetches relevant passages at query time and feeds them into the prompt, so the model answers using information it can actually see.

Why It Matters

Language models are frozen at training time and don't know about a company's internal documents, a user's uploaded files, or anything published after their cutoff date. RAG closes that gap cheaply: no retraining, and the source of an answer can be traced back to a specific chunk of text, which makes it easier to catch hallucinations.

How the Pipeline Works

Indexing: documents are split into chunks, each chunk is converted into a vector embedding, and the embeddings are stored in a vector database alongside the original text.

Retrieval: when a user asks a question, the question itself is embedded and compared against the stored chunk embeddings using a similarity metric such as cosine similarity. The closest matches are pulled back.

Generation: the retrieved chunks are inserted into the prompt as context, and the model is instructed to answer using that context, explicitly saying so if the answer isn't contained in it.

Common Pitfalls

Chunking too coarsely loses precision; chunking too finely loses surrounding context. Retrieving too few chunks can miss the answer; retrieving too many dilutes the prompt and raises cost. And if the underlying documents are stale or wrong, the model will confidently repeat that mistake — RAG grounds answers in a source, but it doesn't guarantee that source is correct.

Beyond Basic RAG

More advanced setups re-rank retrieved chunks before generation, route different question types to different retrieval strategies, or let the model decide when retrieval is even necessary in the first place, rather than running it on every single query.